Generic Apixaban Versus Warfarin for Stroke Prevention in Non-Valvular Atrial Fibrillation

A Cost-Utility and Budget Impact Analysis from the Perspective of the Ontario Public Drug Programs

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About this report. This is an independent portfolio project prepared to demonstrate health economic modelling methods. It is not affiliated with, reviewed by, or endorsed by Canada's Drug Agency (CDA-AMC). It adapts the structure and inputs of the published CADTH worked example [2] and should not be used to inform clinical or funding decisions. Developed with AI assistance under author direction; all analytical choices, inputs, and results reviewed and validated by the author.

Executive Summary

Decision problem. Direct oral anticoagulants (DOACs) were assessed by CADTH in 2012–2013 when they were available only as branded products. Generic apixaban is now listed on the Ontario Drug Benefit (ODB) Formulary at \$0.4084 per 5 mg tablet, 75% below the brand price of \$1.6336. This report asks whether generic entry changes the value of apixaban relative to warfarin for publicly funded patients with non-valvular atrial fibrillation (AF).

Methods. A five-state cohort Markov model with three-month cycles followed patients aged 70 with a $CHADS_2$ score of 2 over a lifetime horizon. Clinical inputs, utilities and event costs were taken from the CADTH 4th-edition guidelines worked example [2]; costs were inflated to 2025 Canadian dollars and drug prices updated to the October 2026 ODB Formulary. The analysis followed the CADTH reference case: public health care payer perspective, 1.5% annual discounting of costs and quality-adjusted life-years (QALYs), and probabilistic analysis with 5,000 simulations.

Results.

- Generic apixaban was **dominant**: it cost \$5,608 less per patient and produced 0.236 more QALYs than warfarin (probabilistic: $-$\$5,589, 95% CI $-$\$8,352 to $-$\$3,242; 0.236 QALYs, 95% CI 0.127 to 0.359).
- Apixaban was cost-saving and more effective in **100%** of 5,000 probabilistic simulations. At the brand price, it was cost-saving in only 13.8%.
- Apixaban remains cost-saving up to **\$1.41 per tablet**, 3.4 times the current generic price, and cost-effective at \$50,000 per QALY up to \$3.51 per tablet.
- The conclusion held in every one-way sensitivity analysis and in all ten scenario analyses, including setting every apixaban treatment effect to the upper bound of its 95% confidence interval.

- Switching an estimated 16,500 remaining warfarin users aged 65+ over three years would cost the drug plan about $6.0 million in drug costs but save the wider health system about $11.3 million.

Conclusion. At current Ontario prices, generic apixaban is expected to be both less costly and more effective than warfarin for stroke prevention in non-valvular AF.

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1 Introduction and Rationale

1.1 Clinical context

Atrial fibrillation is the most common sustained cardiac arrhythmia and increases the risk of ischaemic stroke roughly fivefold. Oral anticoagulation is the mainstay of stroke prevention. For decades the standard option was warfarin, a vitamin K antagonist that is inexpensive to buy but requires regular International Normalized Ratio (INR) blood testing, has many drug and food interactions, and carries a substantial risk of intracranial haemorrhage (ICH).

The DOACs (dabigatran, rivaroxaban, apixaban and edoxaban) were compared with warfarin in large randomised trials. In ARISTOTLE, apixaban 5 mg twice daily reduced stroke or systemic embolism (hazard ratio [HR] 0.79, 95% CI 0.66–0.95), major bleeding (HR 0.69, 95% CI 0.60–0.80), haemorrhagic stroke (HR 0.51) and all-cause death (HR 0.89, 95% CI 0.80–0.99) [5].

1.2 Why revisit the economics now

CADTH's 2012 therapeutic review of the new oral anticoagulants [4] and the economic evaluation underlying it [3] were conducted when apixaban cost about $1,289 per year. At that price, apixaban cost more than warfarin and was judged cost-effective because its health gains justified the additional spending: an incremental cost-effectiveness ratio (ICER) of $22,875 per QALY versus warfarin in patients with a $CHADS_2$ score of 2 [2].

Generic apixaban is now widely available. The ODB Formulary lists interchangeable generic apixaban 5 mg products at $0.4084 per tablet, compared with $1.6336 for brand Eliquis [7]. The annual drug cost has fallen from roughly $1,193 (2026 brand list price) to $298, which is now below the combined annual cost of warfarin and INR monitoring (about $384). When a comparator's price changes this much, the previous economic conclusion needs updating: the question is no longer whether apixaban's benefits justify its extra cost, but whether it still costs extra at all.

1.3 Objectives

1. Estimate the cost-utility of generic apixaban versus warfarin at current Ontario prices, following the CADTH reference case [1].
2. Characterise parameter and structural uncertainty through probabilistic, one-way and scenario analyses.
3. Identify the apixaban price thresholds at which the conclusion would change.
4. Estimate the three-year budget impact of switching remaining warfarin users in Ontario.

2 Methods

2.1 Decision problem

Table 1 summarises the decision problem and the main features of the analysis.

Table 1: Decision problem and reference-case specification

Element	Specification
Population	Adults with non-valvular AF requiring anticoagulation; reference cohort aged 70 with $CHADS_2 = 2$
Intervention	Generic apixaban 5 mg twice daily
Comparator	Warfarin, dose-adjusted to INR 2.0–3.0, with INR monitoring
Outcomes	QALYs, life-years, strokes, ICH, major bleeds; costs in 2025 CAD
Type of evaluation	Cost-utility analysis; supplementary cost comparison and budget impact
Perspective	Publicly funded health care payer (Ontario)
Time horizon	Lifetime (to age 100); scenarios at 2 and 10 years
Discounting	1.5% per year on costs and QALYs; scenarios at 0% and 3%
Model	Cohort Markov model, five health states, three-month cycles
Software	Python 3.11 (NumPy, pandas, Matplotlib)

2.2 Model structure

The model (Figure 1) follows a cohort through five health states. All patients start in **S0** (AF, no prior event), receiving the assigned anticoagulant. In each three-month cycle, patients may die of non-vascular causes or experience one clinical event: stroke, ICH, major extracranial bleed, myocardial infarction (MI), transient ischaemic attack (TIA) or minor bleed. Following the CADTH worked example [2], patients who survive a stroke, ICH or major bleed discontinue anticoagulation and continue on aspirin.

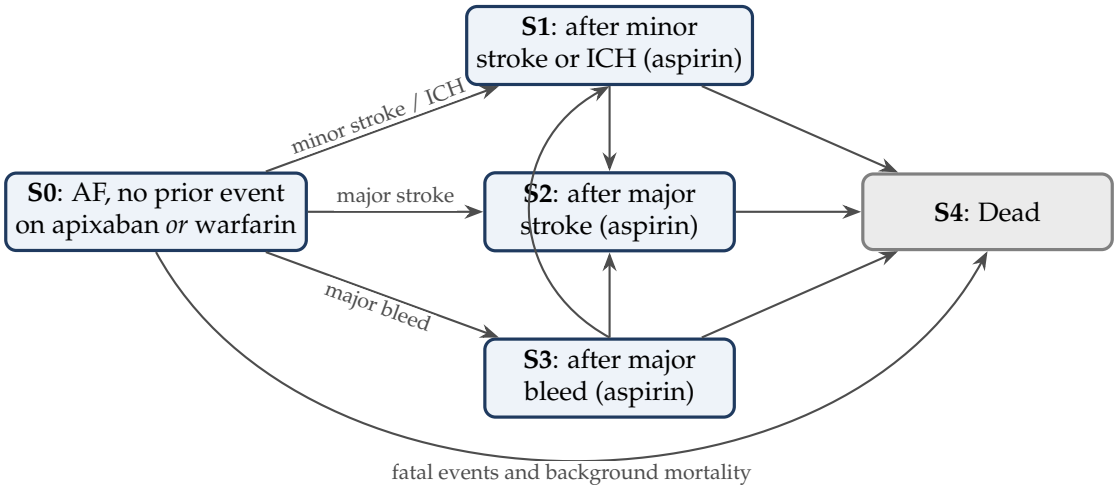


Figure 1: Model structure. Each living state can also remain in place. TIA, minor bleeds and non-fatal MI generate costs and QALY losses in the cycle they occur but do not change state.

The five-state structure is a deliberate simplification of the 173-state model used in the worked example. It keeps the events that drive costs and outcomes in this decision (stroke severity, ICH and major bleeding) while remaining fully auditable. The consequences of this simplification are examined in the validation (Section 3.1).

2.3 Transition probabilities

Event inputs are annual probabilities p observed in the warfarin arm of RE-LY [6] for $CHADS_2 = 2$, as reported in the worked example. Each was converted to a rate, adjusted, and converted back to a probability for the cycle length $\Delta t = 0.25$ years:

$$r = -\ln(1-p), \qquad r' = r \times \mathrm{RR}_{\mathrm{age}} \times \mathrm{RR}_{\mathrm{history}} \times \mathrm{RR}_{\mathrm{aspirin}}, \qquad p_{\mathrm{cycle}} = 1 - e^{-r'\Delta t}. \tag{1}$$

Stroke risk was increased by a factor of 1.5 per decade of age above 70, by 2.2 after a previous stroke or ICH, and by the aspirin-versus-warfarin relative risk in states S1–S3.

The effect of apixaban was applied in S0 only, using odds ratios (OR) from the CADTH network meta-analysis [4]:

$$p_{\mathrm{apixaban}} = \frac{\mathrm{OR} \cdot p_{\mathrm{warfarin}}}{1 - p_{\mathrm{warfarin}} + \mathrm{OR} \cdot p_{\mathrm{warfarin}}}. \tag{2}$$

Odds ratios were 0.79 for stroke, 0.42 for ICH, 0.80 for major bleeding and 0.88 for MI. These are consistent with, and slightly more conservative than, the ARISTOTLE estimates for bleeding [5].

Background (non-vascular) mortality was anchored to the RE-LY rate at age 70 (3.3% per year) and increased with age following the gradient of the Statistics Canada 2021–2023 life table [8]. A Gompertz curve, $\ln q_x = -11.358 + 0.1046x$, was fitted to the life-table probabilities at ages 70–90 to provide a smooth age gradient. Mortality was multiplied by 2.3 after a stroke.

2.4 Health-state utilities

Utilities were taken from the worked example [2]: 0.81 for AF without prior events, 0.75 after a minor stroke or ICH, and a weighted average of 0.39 (modified Rankin 3–4) and 0.11 (Rankin 5) after a major stroke. Utility declined by 0.00029 per year of age above 70. One-off decrements were applied for MI (0.125), major bleed (0.092), TIA (0.103) and minor bleed (0.013) in the cycle of the event.

2.5 Resource use and costs

Costs reflect a public payer perspective in 2025 Canadian dollars (Table 2). Event and long-term health-state costs from the worked example (2011 CAD) were inflated using the all-items Consumer Price Index (164.2/119.9 = 1.3695) [9]. Drug costs used current ODB unit prices without inflation, since generic prices do not track general inflation. Dispensing fees and mark-ups were excluded, consistent with CDA-AMC practice.

Table 2: Key cost inputs (2025 CAD)

Item	Value	Source
Generic apixaban 5 mg twice daily, per year	$298	ODB, Oct 2026 [7]
Brand apixaban (Eliquis) 5 mg twice daily, per year	$1,193	ODB, Oct 2026 [7]
Warfarin 5 mg daily, per year	$55	[2] (2011 ODB)
INR monitoring, per year	$330	[2] $\times$ CPI
Aspirin, per year	$39	[2] (2011 ODB)
Major stroke, acute	$77,874	[2] $\times$ CPI
Minor or fatal stroke, acute	$23,007	[2] $\times$ CPI
Intracranial haemorrhage	$22,677	[2] $\times$ CPI
Major bleed	$6,015	[2] $\times$ CPI
Long-term care after major stroke, per year	$26,115	[2] $\times$ CPI
Long-term care after minor stroke or ICH, per year	$10,813	[2] $\times$ CPI

2.6 Model validation

Following the worked example's validation approach [2], the model was assessed in two ways:

1. **Cross-validation.** Re-running the model with the worked example's 2011 costs and brand apixaban price, and comparing outputs with the published results for $CHADS_2 = 2$.
2. **Internal validity.** (a) Setting all utilities to 1 and removing decrements must make QALYs equal life-years. (b) Setting all apixaban odds ratios to 1 and equalising drug costs must make the two arms identical. (c) Lowering the stroke odds ratio must increase QALYs monotonically. (d) Raising the discount rate must lower discounted QALYs. Mass balance (state occupancy summing to 1) was checked in every cycle.

2.7 Analysis of uncertainty

Probabilistic analysis. Following the worked example's conventions, probabilities and utilities were assigned beta distributions, odds ratios and relative risks lognormal distributions based on their 95% confidence intervals, event costs gamma distributions with a 25% coefficient of variation, and utility decrements normal distributions. Drug prices were fixed. Expected values were estimated from 5,000 Monte Carlo simulations with a fixed random seed. The same simulations were repeated at the brand apixaban price to isolate the effect of generic entry.

One-way sensitivity analysis. Each of the 50 parameters was set to its low and high values (Appendix A) while all others were held at base values. Results are reported as incremental net monetary benefit (INMB) at $50,000 per QALY, since ICERs are not meaningful when one option dominates.

Scenario analyses. Scenarios addressed methodological and structural uncertainty: discount rates of 0% and 3%, time horizons of 2 and 10 years, waning of the apixaban effect by 5% per year after year 2 (applied to the log odds ratios), all apixaban odds ratios at the upper bound of their 95% confidence intervals, no stroke benefit, no INR monitoring cost, a cohort starting at age 80, and the brand apixaban price.

Price threshold analysis. Because incremental cost is linear in the apixaban price, the prices at which apixaban becomes cost-neutral and at which the ICER equals $50,000 per QALY were solved exactly.

2.8 Cost comparison and budget impact

A cost comparison of oral anticoagulants at current ODB prices was prepared to show the drug acquisition costs a payer faces. A three-year budget impact analysis estimated the cost of switching Ontario residents aged 65 and over who remain on warfarin to generic apixaban. The eligible population was derived from the Ontario population aged 65+ (3.1 million in 2025 [10]), AF prevalence of 5.9% [11], 60% treated with an oral anticoagulant, and an assumed 15% of treated patients still on warfarin. Uptake was assumed to be 25%, 50% and 75% in years 1 to 3. Results are presented from two perspectives: the drug plan (drug acquisition costs only, the CDA-AMC reference case for budget impact) and the wider health system (drugs, monitoring, events and long-term care, from the Markov model).

3 Results

3.1 Validation

All four internal validity checks passed, and state occupancy summed to 1 in all 120 cycles. Table 3 compares the model with the published worked example at 2011 prices.

Table 3: Cross-validation against the CADTH worked example (2011 prices, brand apixaban, $CHADS_2 = 2$)

Outcome	CADTH (2017)	This model	Difference
Warfarin: total cost	\$20,558	\$20,875	+1.5%
Warfarin: life-years (discounted)	9.12	9.13	+0.1%
Warfarin: QALYs (discounted)	7.49	7.23	−3.5%
Apixaban: total cost	\$24,665	\$24,919	+1.0%
Apixaban: life-years (discounted)	9.48	9.39	−1.0%
Apixaban: QALYs (discounted)	7.67	7.46	−2.7%
Incremental cost	\$4,107	\$4,044	−1.5%
Incremental QALYs	0.180	0.236	+31.3%
ICER (\$ per QALY)	\$22,875	\$17,117	−25.2%
Strokes per 1,000 (warfarin / apixaban)	242 / 201	282 / 239	
ICH per 1,000 (warfarin / apixaban)	61 / 31	70 / 36	
Major bleeds per 1,000 (warfarin / apixaban)	298 / 235	339 / 292	

Total costs, life-years and incremental costs were reproduced within 2%, and the absolute number of events avoided was similar (43 strokes and 34 ICH avoided per 1,000 patients, against 41 and 30 in the worked example). However, the model predicted 14–24% more events in both arms, and its incremental QALY gain was 31% larger, giving an ICER 25% lower than the published value. Likely contributors are the simpler state structure (no tracking of prior TIA, MI or pulmonary embolism), the comparison of deterministic results with published probabilistic means, and the continuous application of the age effect on stroke risk. Both models reach the same decision at 2011 prices. Because this model appears somewhat favourable to apixaban, the scenario analyses deliberately include pessimistic treatment-effect assumptions.

3.2 Reference case

Table 4 presents the deterministic reference-case results. Generic apixaban was less costly (\$22,808 versus \$28,416) and more effective (7.464 versus 7.228 QALYs) than warfarin, and is therefore

dominant. The incremental net monetary benefit at $50,000 per QALY was $17,421 per patient.

Table 4: Reference-case results per patient (deterministic, 2025 CAD)

	Generic apixaban	Warfarin	Incremental
Discounted			
Total cost	$22,808	$28,416	−$5,608
Drug and monitoring	$2,362	$2,791	−$429
Acute clinical events	$11,500	$13,953	−$2,453
Long-term post-event care	$8,946	$11,672	−$2,726
QALYs	7.464	7.228	0.236
Life-years	9.388	9.129	0.258
Undiscounted			
Total cost	$25,757	$31,952	−$6,194
QALYs	8.225	7.944	0.281
Life-years	10.352	10.043	0.309
Clinical events per 1,000 patients (lifetime)			
Strokes	238.6	281.8	−43.1
Intracranial haemorrhages	35.5	69.8	−34.3
Major bleeds	292.4	338.9	−46.6
ICER	Apixaban dominant (less costly, more effective)		
INMB at $50,000 per QALY			$17,421

Only 8% of the cost saving came from lower drug and monitoring costs. The rest came from avoided acute events (44%) and avoided long-term care after stroke (49%), so the result depends mainly on apixaban's clinical benefit, particularly its halving of ICH. Figure 2 shows the cohort's distribution across health states over time.

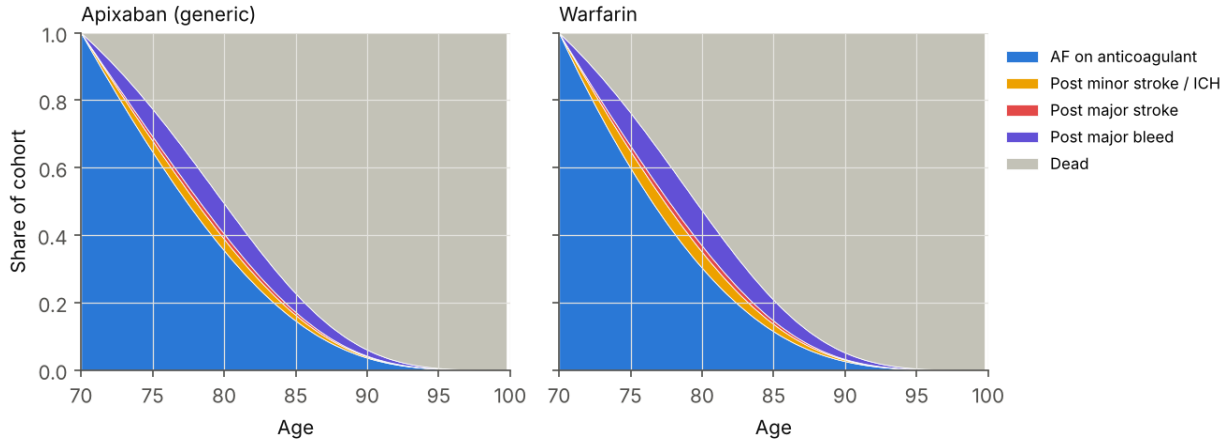


Figure 2: Markov trace: share of the cohort in each health state by age.

3.3 Price threshold analysis

Each $0.10 increase in the price per tablet raised the incremental cost by $562 per patient (Figure 3). Apixaban remained cost-saving up to **$1.4059 per tablet**, 3.4 times the current generic price, and cost-effective at $50,000 per QALY up to **$3.5072 per tablet**. At the brand price of $1.6336, apixaban

would cost $1,280 more per patient with an ICER of $5,419 per QALY: still good value, but no longer cost-saving.

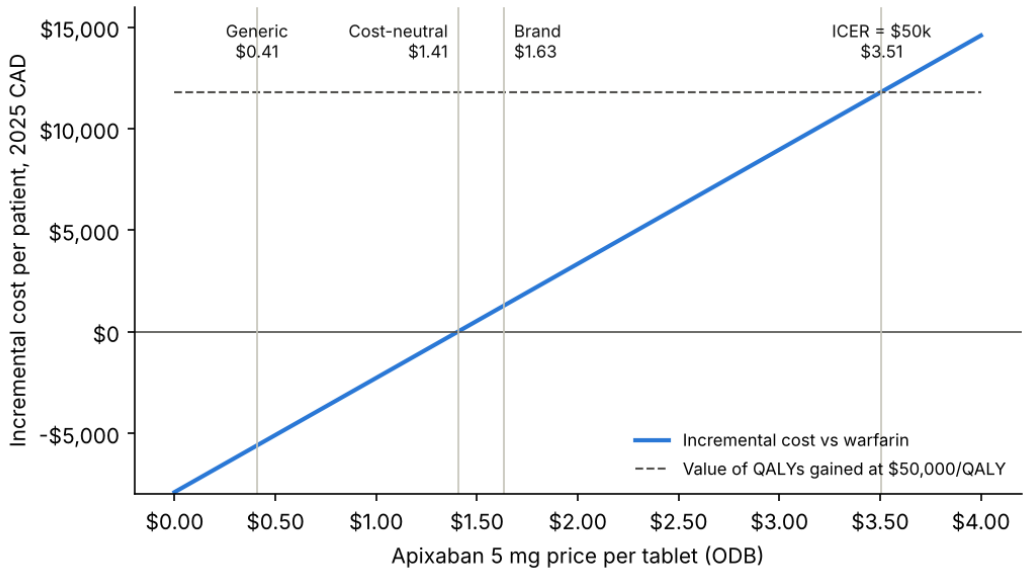


Figure 3: Incremental cost of apixaban versus warfarin as a function of the price per tablet.

3.4 Probabilistic analysis

Across 5,000 simulations, the mean incremental cost was $-\$5,589$ (95% CI $-\$8,352$ to $-\$3,242$) and the mean incremental QALY gain was 0.236 (95% CI 0.127 to 0.359). Generic apixaban was less costly and more effective in all 5,000 simulations (Figure 4), so the probability that it is cost-effective is 100% at every willingness-to-pay threshold. The running mean of incremental cost changed by less than 1% between 1,000 and 5,000 simulations ($-\$5,629$ versus $-\$5,589$), indicating convergence.

Repeating the same 5,000 simulations at the brand price gave a mean incremental cost of $1,292 (95% CI $-\$1,345$ to $3,499). Apixaban was then cost-saving in only 13.8% of simulations, and the probability of being cost-effective rose with willingness to pay: 45% at $5,000, 74% at $10,000, 95% at $20,000 and 99.7% at $50,000 per QALY (Figure 5).

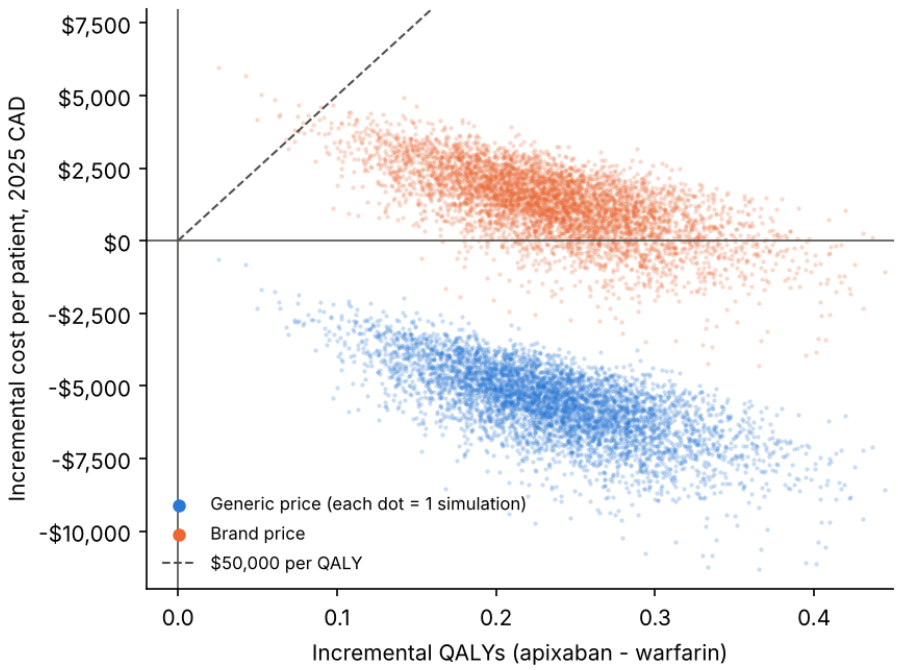


Figure 4: Cost-effectiveness plane: 5,000 probabilistic simulations at the generic and brand apixaban prices.

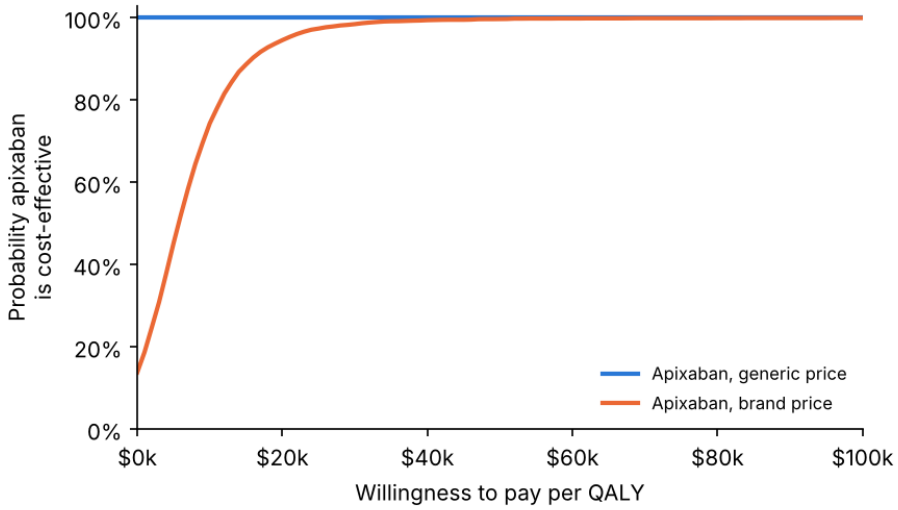


Figure 5: Cost-effectiveness acceptability curves at the generic and brand apixaban prices.

3.5 One-way sensitivity analysis

The most influential parameters were the apixaban odds ratio for stroke, the apixaban price and background mortality (Figure 6). No parameter, at any value within its range, made the INMB negative; the lowest value was $10,533, at the brand price.

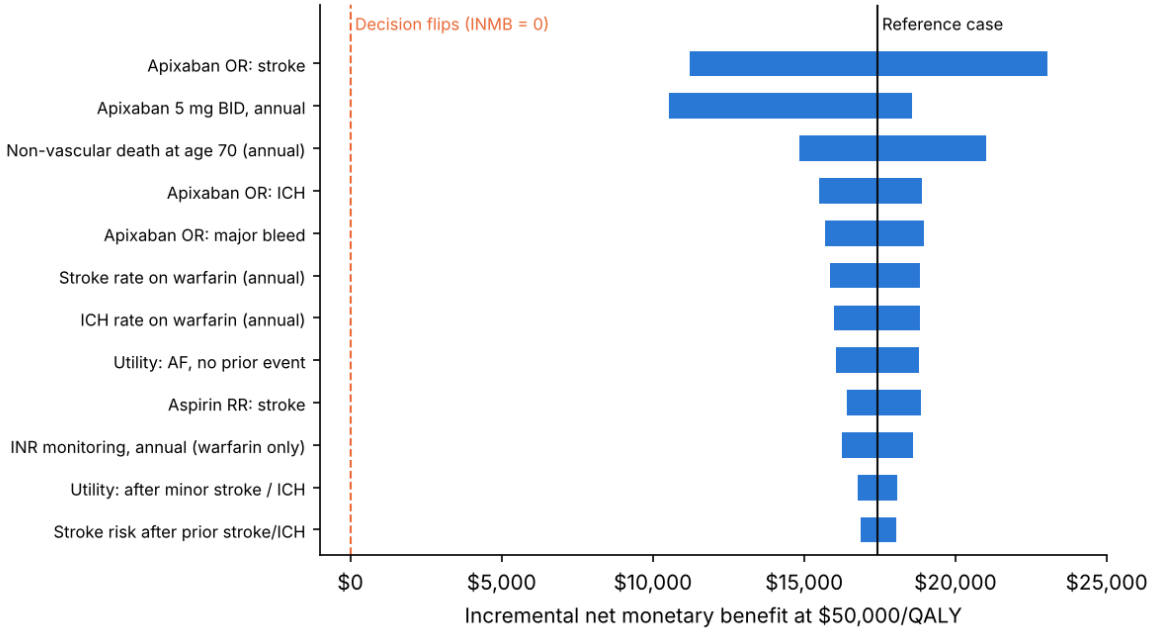


Figure 6: Tornado diagram: incremental net monetary benefit at $50,000 per QALY for the 12 most influential parameters.

3.6 Scenario analyses

Generic apixaban remained dominant in all scenarios (Table 5). The most demanding test, setting every apixaban odds ratio to the upper bound of its 95% confidence interval, reduced the QALY gain from 0.236 to 0.092 but apixaban still saved $2,614 per patient. This scenario reduces the treatment benefit far more than the 31% difference in incremental QALYs found in the validation. Even within a two-year horizon, the length of the pivotal trials, apixaban was cost-saving.

Table 5: Scenario analyses (deterministic, per patient)

Scenario	Incr. cost	Incr. QALYs	Result	INMB
Reference case	−$5,608	0.236	Dominant	$17,421
Brand apixaban price (Eliquis)	$1,280	0.236	$5,419/QALY	$10,533
Discount rate 0%	−$6,194	0.281	Dominant	$20,222
Discount rate 3%	−$5,105	0.201	Dominant	$15,137
Time horizon 2 years	−$822	0.004	Dominant	$1,032
Time horizon 10 years	−$4,357	0.097	Dominant	$9,231
Effect wanes 5%/year after year 2	−$4,912	0.207	Dominant	$15,257
All apixaban ORs at upper 95% CI	−$2,614	0.092	Dominant	$7,238
No stroke benefit (OR = 1)	−$3,051	0.125	Dominant	$9,306
No INR monitoring cost	−$3,283	0.236	Dominant	$15,096
Cohort starts at age 80	−$3,125	0.105	Dominant	$8,384

3.7 Cost comparison

At current ODB prices, the annual drug cost of generic apixaban ($298) is lower than warfarin with INR monitoring ($384) and more than 75% below the brand price (Table 6). Generic edoxaban

60 mg once daily costs slightly less again ($268 per year). Whether the two generic DOACs can be treated as clinically similar is a question for a clinical review and was not assessed here; edoxaban's price is shown for context.

Table 6: Drug acquisition costs of oral anticoagulants, ODB Formulary, October 2026

Regimen	Dosing	Unit price	Daily cost	Annual cost
Apixaban 5 mg, generic	1 tablet twice daily	$0.4084	$0.82	$298
Edoxaban 60 mg, generic	1 tablet once daily	$0.7348	$0.73	$268
Apixaban 5 mg, brand (Eliquis)	1 tablet twice daily	$1.6336	$3.27	$1,193
Warfarin 5 mg + INR monitoring	daily, with INR testing	–	$1.05	$384

3.8 Budget impact

An estimated 16,461 Ontarians aged 65+ remain on warfarin under the base-case assumptions. Switching them to generic apixaban at the assumed uptake would increase drug-plan spending by $1.0 million in year 1, $2.0 million in year 2 and $3.0 million in year 3 ($6.0 million in total), since generic apixaban costs $244 more per patient-year than warfarin tablets alone (Table 7). From the wider health-system perspective, which includes INR monitoring, avoided events and long-term care, the switch would save $1.6 million, $3.7 million and $6.0 million in years 1 to 3 ($11.3 million in total).

Table 7: Three-year budget impact of switching remaining warfarin users aged 65+ to generic apixaban, Ontario

	Year 1	Year 2	Year 3	Total
Patients switched	4,115	8,231	12,346	
Drug plan (drug acquisition only)	$1.00M	$2.01M	$3.01M	$6.02M
Health system (all costs)	−$1.60M	−$3.66M	−$5.99M	−$11.25M

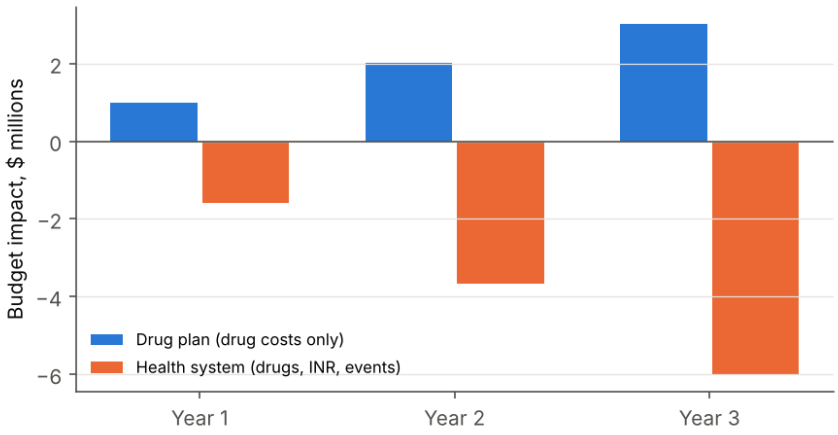


Figure 7: Budget impact by year and perspective.

The drug-plan estimate is most sensitive to the share of patients still on warfarin, for which no current Ontario estimate was found: at 5% the three-year cost would be $2.0 million, and at 30% it

would be $12.0 million.

4 Discussion

4.1 Interpretation

When CADTH evaluated the DOACs, apixaban offered better outcomes at a higher cost and was judged good value at about $23,000 per QALY. With generic entry, that trade-off has disappeared: apixaban now costs less than warfarin even before its clinical benefits are counted, and its benefits generate further savings by preventing strokes and intracranial haemorrhages that are expensive to treat and to live with. Apixaban was dominant in every analysis performed, including deliberately pessimistic ones.

The budget impact illustrates a familiar tension in pharmaceutical policy. The drug plan would pay more for anticoagulant tablets, while the savings accrue to hospitals, laboratories and long-term care, which are funded from different budgets. A decision-maker focused on the drug budget alone would see a cost; a decision-maker considering the whole public health system would see a saving.

4.2 Limitations

- **Simplified structure.** The five-state model omits history of TIA, MI and pulmonary embolism. Validation showed it reproduces costs and life-years well but overstates the incremental QALY gain relative to the worked example. Pessimistic scenarios were included to address this, and the conclusion did not change.
- **Age of the evidence.** The odds ratios come from a 2012 network meta-analysis, and utilities and event costs date from studies published in 1996–2012. These remain the inputs used in CADTH's published worked example, but a full review would update them.
- **Single subgroup.** Only $CHADS_2$ = 2 at age 70 was modelled, with an age-80 scenario. Higher-risk patients would be expected to gain more from anticoagulation.
- **Warfarin price.** The warfarin drug cost is the 2011 ODB value; the current listing could not be verified. Warfarin is a small share of total costs and the conclusion is not sensitive to it.
- **Comparators.** Rivaroxaban and dabigatran were not modelled because their current ODB prices could not be verified, and no sequential analysis across DOACs was performed.
- **Adherence and dosing.** The model does not include treatment discontinuation, reduced apixaban doses (2.5 mg is priced identically), or real-world INR control.
- **Budget impact assumptions.** AF prevalence, treatment rates, the share of patients on warfarin and uptake are assumptions, and health-system savings were estimated as if each cohort started therapy at age 70.

5 Conclusion

At current Ontario Drug Benefit prices, generic apixaban is expected to be less costly and more effective than warfarin for stroke prevention in non-valvular atrial fibrillation. The result is robust to parameter uncertainty, to pessimistic assumptions about apixaban's treatment effect and to alternative time horizons and discount rates, and would hold unless the price rose above about

$1.41 per tablet. Switching remaining warfarin users would modestly increase drug-plan spending while reducing total health-system costs.

Reproducibility

The model is implemented in Python as a series of annotated Jupyter notebooks covering inputs, the Markov engine, validation, reference case, probabilistic analysis and sensitivity analysis. Each notebook reads the same input file, and all results in this report can be regenerated from the code. Repository: `https://github.com/98DanPazos/doac-cua-ontario`.

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A Model Parameters

Table 8: All model parameters: base values, one-way sensitivity ranges and probabilistic distributions. Costs in 2025 CAD. LN = lognormal, parameterised from the 95% CI shown; Beta n = effective sample size; CV = coefficient of variation; SE = standard error.

Parameter	Base	Range (OWSA)	PSA distribution	Source
Stroke rate on warfarin (annual)	0.016	0.012–0.02	Beta (n = 11,794)	[2]
TIA rate (annual)	0.008	0.006–0.01	Beta (n = 11,794)	[2]
ICH rate on warfarin (annual)	0.008	0.006–0.01	Beta (n = 11,794)	[2]
Major bleed rate on warfarin (annual)	0.033	0.025–0.041	Beta (n = 11,794)	[2]
Minor bleed rate (annual)	0.164	0.123–0.205	Beta (n = 11,794)	[2]
MI rate (annual)	0.006	0.0045–0.0075	Beta (n = 11,794)	[2]
Proportion of first strokes fatal	0.237	0.18–0.3	Beta (n = 186)	[2]
Proportion of non-fatal strokes major	0.333	0.25–0.42	Beta (n = 117)	[2]
Increase in fatality of recurrent stroke	1.57	1.21–2.03	LN (1.21–2.03)	[2]
Proportion of major bleeds / ICH fatal	0.084	0.06–0.11	Beta (n = 476)	[2]
Proportion of MIs fatal	0.121	0.06–0.2	Beta (n = 66)	[2]
Stroke risk per 10 years of age	1.5	1.3–1.7	LN (1.3–1.7)	[2]
Stroke risk after prior stroke/ICH	2.2	1.2–3.5	LN (0.78–6.35)	[2]
Mortality after stroke	2.3	2–2.7	LN (2–2.7)	[2]
Non-vascular death at age 70 (annual)	0.033	0.025–0.041	Beta (n = 11,794)	[2]
Apixaban OR: stroke	0.79	0.65–0.95	LN (0.65–0.95)	[2]
Apixaban OR: MI	0.88	0.66–1.17	LN (0.66–1.17)	[2]
Apixaban OR: ICH	0.42	0.3–0.58	LN (0.3–0.58)	[2]
Apixaban OR: major bleed	0.8	0.68–0.94	LN (0.68–0.94)	[2]
Aspirin RR: stroke	1.62	0.99–2.65	LN (0.99–2.65)	[2]
Aspirin RR: MI	1.42	0.84–2.39	LN (0.84–2.39)	[2]
Aspirin RR: ICH	0.51	0.16–1.6	LN (0.16–1.6)	[2]
Aspirin RR: minor bleed	0.63	0.32–1.22	LN (0.32–1.22)	[2]
Aspirin RR: major bleed	1.14	0.47–2.73	LN (0.47–2.73)	[2]
Aspirin RR: TIA	1.56	0.86–2.83	LN (0.86–2.83)	[2]
Utility: AF, no prior event	0.81	0.75–0.87	Beta (n = 42)	[2]
Utility: after minor stroke / ICH	0.75	0.67–0.83	Beta (n = 116)	[2]
Utility: major stroke, Rankin 3-4	0.39	0.32–0.46	Beta (n = 179)	[2]
Utility: major stroke, Rankin 5	0.11	0.06–0.16	Beta (n = 172)	[2]
Share of major strokes at Rankin 5	0.205	0.1–0.31	Beta (n = 47)	[2]
Utility loss per year over 70	0.00029	0.00025–0.00033	Normal (SE 0.00002)	[2]
Disutility: MI	0.125	0.107–0.143	Normal (SE 0.009)	[2]
Disutility: major bleed	0.092	0.072–0.112	Normal (SE 0.01)	[2]
Disutility: minor bleed	0.013	0.011–0.015	Normal (SE 0.001)	[2]
Disutility: TIA	0.103	0.087–0.119	Normal (SE 0.008)	[2]
Cost: fatal stroke	$23,007	$17,255–$28,759	Gamma (CV 25%)	[2] × CPI
Cost: minor stroke	$23,007	$17,255–$28,759	Gamma (CV 25%)	[2] × CPI
Cost: major stroke (acute)	$77,874	$58,405–$97,342	Gamma (CV 25%)	[2] × CPI
Cost: TIA	$5,883	$4,412–$7,354	Gamma (CV 25%)	[2] × CPI
Cost: ICH	$22,677	$17,008–$28,346	Gamma (CV 25%)	[2] × CPI
Cost: major bleed	$6,015	$4,511–$7,518	Gamma (CV 25%)	[2] × CPI
Cost: minor bleed	$142	$107–$178	Gamma (CV 25%)	[2] × CPI
Cost: fatal MI	$10,067	$7,550–$12,584	Gamma (CV 25%)	[2] × CPI
Cost: non-fatal MI	$15,585	$11,688–$19,481	Gamma (CV 25%)	[2] × CPI
Annual cost: after major stroke	$26,115	$19,586–$32,643	Gamma (CV 25%)	[2] × CPI
Annual cost: after minor stroke / ICH	$10,813	$8,110–$13,517	Gamma (CV 25%)	[2] × CPI
Apixaban 5 mg BID, annual	$298	$149–$1,193	Fixed	ODB 2026
Warfarin 5 mg daily, annual	$54.61	$27.00–$110	Fixed	[2] (2011 ODB)
INR monitoring, annual (warfarin only)	$330	$165–$494	Fixed	[2] × CPI
Aspirin, annual	$39.04	$20.00–$60.00	Fixed	[2] (2011 ODB)